

Does the Conservative Formula still work?

Replication design and extensions

MTH 9897 / Project 8

First-draft presentation

Synthetic demonstration results

Research question

Does combining stable stocks with momentum and shareholder payouts improve performance?

Comparisons

Market benchmark
Single-signal strategies
Results after trading costs

Current status

Executable research prototype
Historical equity data pending
Empirical conclusions remain open

The portfolio selection rule

Paper recipe at each quarter end, with sufficient eligible history

1,000

Largest eligible US stocks by market capitalization

500

Lower-volatility half using 36 months of returns

100

Best average ranks on momentum and payout yield

Equal weights at formation. Quarterly rebalancing. Historical sample counts may be smaller.

Signal definitions in the draft

Signal	Draft construction	Preference
Volatility	36-month return standard deviation	Lower
Momentum	11-month price return, skipping latest month	Higher
Payout proxy	Dividend yield + 1 – shares / 24-month mean	Higher

Payout definition remains subject to reconciliation with the paper.

Information timing

Example: portfolio formed at December month end

Input or action	Window / timing
Volatility	36 months ending in December
Momentum	January through November
Selection	December month-end information
First earned return	Following January

Quarter-end execution is idealized. Late-reported inputs require an additional lag.

Historical study and demonstration data

Dimension	Intended empirical study	Current demonstration
Source	WRDS / CRSP	Seeded synthetic generator
Return period	1929 to latest full year	January 2003–December 2025
Universe	Historical eligible US stocks	1,100 artificial securities
Interpretation	Test the paper's findings	Illustrate the research workflow

Historical work must retain delisted names and reconcile legacy versus CIZ return conventions.

Portfolio accounting

Quarterly targets

Weights drift between rebalances

Trading costs

Charge both purchases and sales

Terminal events

Final proceeds move to cash

Missing held returns

Stop and reconcile the data

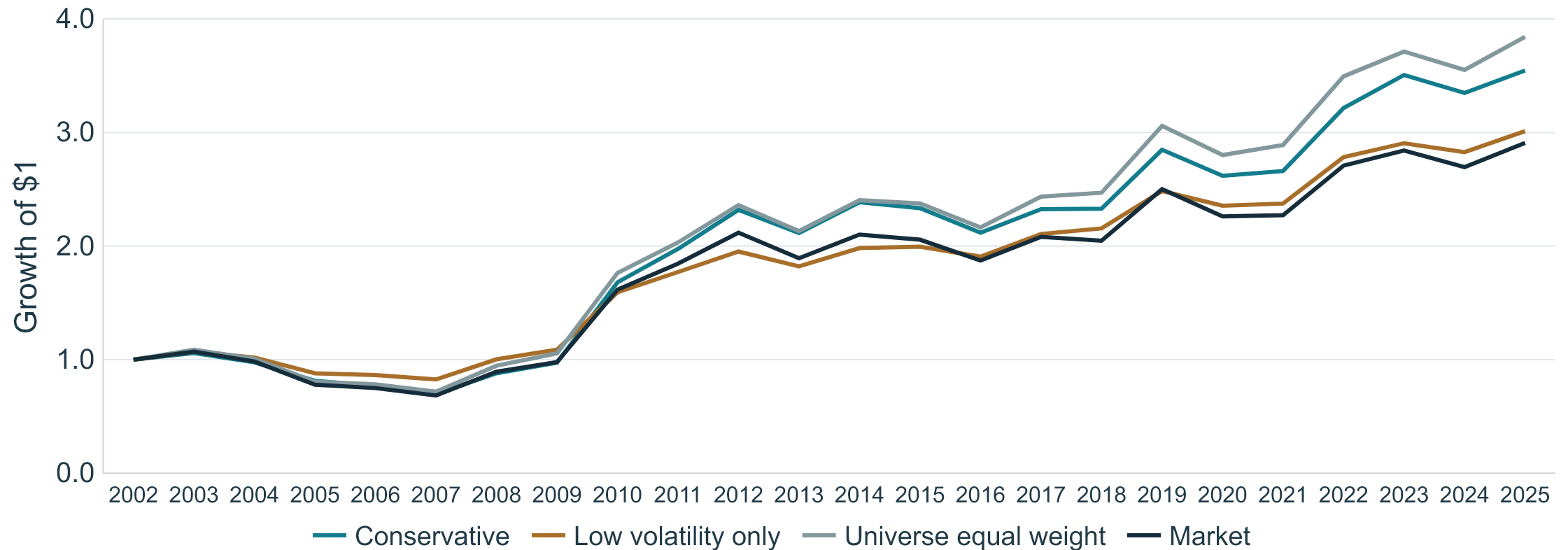
Costs, cash returns, and initial purchases enter the backtest explicitly.

Experiments

Comparison	Research purpose
Remove momentum or payout	Measure each signal's incremental contribution
Remove the volatility screen	Test whether defensive selection matters
Equal / cap-weight universe	Separate selection effects from weighting
Later periods and costs	Assess stability and implementation sensitivity

Demonstration wealth paths

SYNTHETIC DEMONSTRATION



Year-end wealth, 2002–2025. Strategies include 10 bps per dollar traded. Synthetic market is gross.

Demonstration performance measures

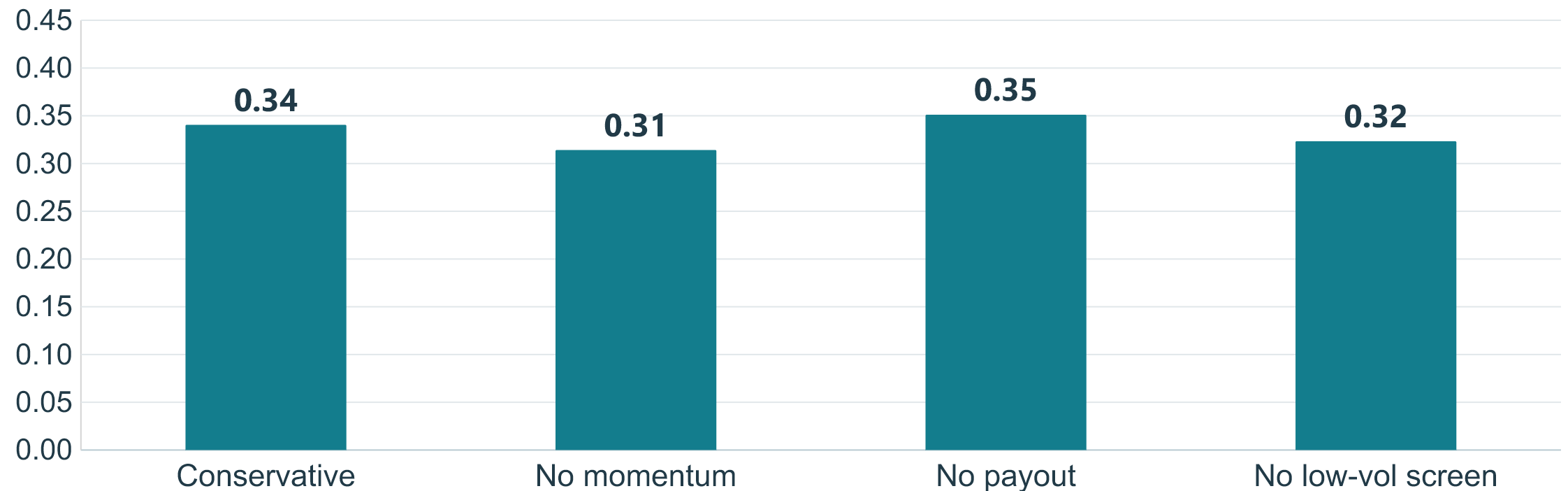
SYNTHETIC DEMONSTRATION

Portfolio	CAGR	Volatility	Sharpe	Max drawdown
Conservative	5.66%	13.61%	0.34	-38.28%
Low volatility only	4.91%	9.99%	0.35	-28.22%
Universe equal weight	6.03%	14.08%	0.36	-38.84%
Market	4.75%	14.16%	0.27	-41.06%

January 2003–December 2025. Synthetic results provide no evidence of real-world outperformance.

Synthetic signal comparisons

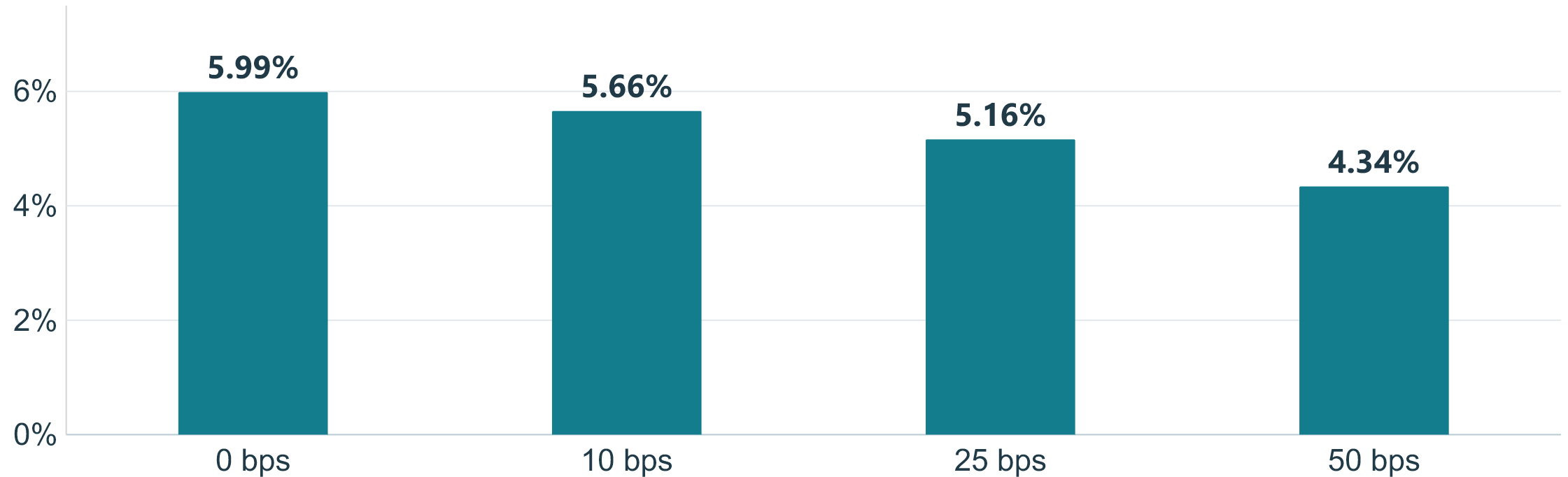
SYNTHETIC DEMONSTRATION



Annualized Sharpe, January 2003–December 2025. Interpretation awaits historical equity data.

Synthetic trading-cost sensitivity

SYNTHETIC DEMONSTRATION



Synthetic CAGR, 2003–2025. Basis points apply to each dollar traded. These are assumed cost scenarios.

Empirical conclusions remain open

The draft defines a reproducible test of the Conservative Formula

Remaining work

Historical CRSP data and corporate-action audit
Payout definition and sample reconciliation
Empirical comparisons and uncertainty estimates

A successful replication may confirm, weaken, or reject the original findings.

References and project files

Van Vliet and Blitz (2018)

The Conservative Formula: Quantitative Investing Made Easy
SSRN 3145152

Supporting sources

Robeco, Guide to Conservative Investing (2023)
CRSP, Summary of CIZ Differences to Legacy Files
MTH 9897, Final Projects 2026

Notebook, slides, and generator code

github.com/yingnichang/MTH9897/tree/main/Final